Anup Chavan

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.64
Millennium PU College Kalaburagi	Class XII (KSEAB)	2023	92.00%
Appa Public School Kalaburagi	Class X (CBSE)	2020	95.00%

WORK EXPERIENCE

Center for Cryptography and Cybersecurity, IIT Hyderabad

Summer 2026

- Prototyped wired network access control for a 90+ workstation lab using openNDS and FreeRADIUS.
- Eliminated post-login VLAN changes and DHCP renewals through gateway-based session enforcement.
- Configured fail-closed firewall rules and switch ACLs to restrict unauthenticated and lateral traffic.

PROJECTS

Obsidian Custom Views | TypeScript, CodeMirror 6, Astro

Jan 2026 - Present

GitHub · Website

- Reached **2,900+ downloads and 89 GitHub stars** with a customizable Markdown-view plugin.
- Engineered a rule-based template engine with loops, conditionals, scoped CSS, and editable note views.
- Designed a live template editor and community gallery with previews, downloads, and submission review.

Kindred: Movie Discovery | Python, NumPy, BGE

Aug 2026 - Present

GitHub · Website

- Indexed **16,284 films** from subtitle-derived profiles, encoding **6 narrative facets** as 768-dimensional vectors.
- Built weighted cosine retrieval with story/feeling controls and emotional-arc reranking.
- Precomputed embeddings at ingestion to serve recommendations without per-query LLM calls or retraining.

FLAKE: Peer-to-Peer Federated Learning | Python, PyTorch

May 2026

GitHub · Website

- Implemented **2 serverless learning protocols** over TCP for clients with non-IID CIFAR-10 data.
- Combined peer-weight aggregation and knowledge distillation using cross-entropy and KL-divergence loss.
- Built reproducible comparisons with shared seeds, identical data splits, and matched model initialization.

gemOS Virtual Memory | C, x86-64 | OS coursework

Apr 2025

- Implemented mmap, munmap, and mprotect with virtual-memory area splitting, merging, and access checks.
- Added demand paging through page-fault handling and four-level page-table allocation.
- Managed physical-frame mappings and TLB invalidation during memory unmapping and protection changes.

RISC-V Assembler and Simulator | C | Team project

Aug 2024 - Nov 2024

GitHub · Website

- Implemented RV64I assembly translation and instruction execution for arithmetic, branches, and memory access.
- Built a configurable data-cache simulator with **3 replacement policies**: LRU, FIFO, and random.
- Added stepping, breakpoints, and cache statistics; tested arithmetic, branching, Fibonacci, and load/store.

Infinite Precision Arithmetic | C++ | Team project

Apr 2024

GitHub · Website

- Co-developed Integer and Float classes using string representations for arithmetic beyond fixed-width integers.
- Implemented overloaded addition, subtraction, multiplication, and division with sign and decimal handling.
- Added input validation, a command-line calculator, and a Makefile target for building a reusable static library.

RELEVANT COURSES

- **Core CS:** Data Structures, Algorithms, Operating Systems, DBMS, Computer Architecture, Software Engineering.
- **Systems:** Parallel and Concurrent Programming, Computer Networks, Compilers-II, Secure Software Engineering.
- **AI/ML:** Data Mining, Foundations of Machine Learning, Distributed AI Training (Large-Scale GPU Clusters).
- **Mathematics:** Discrete Mathematics, Linear Algebra, Probability and Random Variables, Introduction to Statistics.
- **Supporting coursework:** Communication Skills, Introduction to Entrepreneurship, Ethics and Values.

SKILLS

- **Languages:** C, C++, Python, Rust, TypeScript, JavaScript, SQL.
- **Engineering:** React, Node.js, Astro, PyTorch, PostgreSQL, Supabase, Git, Docker, Linux.
- **Design:** UI/UX design, interaction design, responsive interfaces, HTML/CSS.